# The Central Distribution: A Noninformative Prior for Feature Selection

Behrouz Ayati
aybehrouz@gmail.com

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# 1 Introduction

## 1.1 The Optimal Strategy

We begin by revisiting a fundamental decision-making problem, and how it is treated within the framework of Bayesian decision theory.

Consider an environment where an agent must select an action from a set of available actions. The loss associated with each action depends on the unknown state of the environment and can be quantified by a real number. Specifically, let $a$ denote an action, and $x$ represent the unknown environment state. The loss function $\ell(a, x)$ quantifies the loss for each action.

Furthermore, we assume that an evidence $e$ of this unknown environment state exists, and the agent can leverage this evidence to determine the best action. In other words, our agent aims to find an optimal strategy, where a strategy defines the distribution of each action conditioned on the observed evidence: $S(A \mid E)$.

According to Bayesian decision theory, the optimal strategy is the one that minimizes the expected loss:

$$S_{\text{opt}} = \arg\min_{S} \sum_{a} \sum_{x,e} \ell(a, x) S(a \mid e) P^*(x, e),$$

where $P^*$ is the joint probability distribution of the environment state and the evidence.

It can be shown that there always exists a deterministic optimal strategy. That is, we can consider a strategy as a function of the evidence, rather than as a conditional distribution:

$$S_{\text{opt}} = \arg\min_S \sum_{x,e} \ell(S(e), x) P^*(x, e).$$

A strategy that, for each evidence $e$, selects the action minimizing the expected loss conditioned on $e$, is optimal:

$$\begin{aligned} S_{\text{opt}}(e) &= \arg\min_a \mathbb{E}_{P^*}\left[\ell(a, x) \mid e\right] \\ &= \arg\min_a \sum_x \ell(a, x) P^*(x \mid e)\cdot \end{aligned} \tag{1}$$

Now, suppose that we decide to use a different evidence $E'$ instead of $E$. Could this change potentially improve our strategy? Within the framework of Bayesian decision theory, we can easily answer this question: To compare any two strategies, we can simply compare their expected loss. The strategy with the lower expected loss is the better one.

Note that, in reality, we cannot freely choose any random variable as our evidence. We can only choose evidence that is a deterministic function of the evidence provided by the environment, allowing us to determine its value after observing the original evidence. Utilizing a strategy that uses a simpler evidence, resembles feature selection in machine learning. For instance, in a learning task where our evidence is an image, we might substitute the image with the first three coefficients of its Fourier transform. Naturally, these coefficients are a function of the original evidence.

Interestingly, if we assume $E' = f(E)$, it turns out that, according to our Bayesian framework, the optimal strategy for $E$ is always better than any strategy that uses $E'$. Thus, at first glance, it appears that Bayesian decision theory does not favor feature selection.

From a certain perspective, it could be argued that adopting a strategy which uses simpler evidence may lead to ignoring some parts of useful available information, and therefore is not justified by Bayesian decision theory.

The key point here is that the space of strategies using $E'$ is a subset of the strategies that use $E$. When we are searching for the best strategy based on $E$, we are implicitly evaluating all simpler forms as well.

It is possible that this explains why the Bayesian framework does not favor the simpler evidence $E'$. Plausibly, if we choose an appropriate prior distribution, when we are finding the optimal strategy, feature selection should be performed automatically within the Bayesian framework. However, many commonly used noninformative priors, such as the flat prior, do not exhibit feature selection characteristics.

Throughout this discussion, we will explore a systematic approach for finding a noninformative prior that also exhibit feature selection characteristics.

## 1.2 Divergence

Suppose that for a given environment, we have specified a probability measure $P$. In particular, we are interested in the distribution it induces on a random variable $X$, denoted $P(X)$.

Now, imagine we ask an expert—a person familiar with the environment—to evaluate our choice of $P$. The expert may hold different beliefs and instead favor another probability measure $Q$. From the expert's perspective, evaluating $P$ involves assessing the potential loss incurred by using the (in their view) incorrect distribution $P(X)$ when the true distribution is believed to be $Q(X)$.

Let us assume that a real number can be used to measure this loss, and denote it as $\mathbb{D}\left[Q(X) \,\|\, P(X)\right]$. It seems reasonable, as $P$ gets closer to $Q$ this measure decreases, and increases when $P$ diverges from $Q$. Therefore, $\mathbb{D}$ effectively is a measure of the divergence of $P(X)$ from $Q(X)$, when $Q$ is the believed true probability measure.

Interestingly, if we make certain assumptions about the properties of this divergence measure, it turns out that—up to a constant factor—it must be equal to the Kullback–Leibler (KL) divergence.

Alternatively, by adopting a more minimal set of properties, we can prove that if we accept Shannon's entropy as a measure of the uncertainty of a distribution, we should also consider KL divergence as the appropriate method for measuring divergence between distributions.

**Theorem 1.** *Let $X$ be a discrete random variable. Suppose $\mathbb{D}\left[Q(X) \,\|\, P(X)\right]$ is some divergence measure that quantifies the divergence of $P(X)$ from $Q(X)$. Consider the following properties:*

**Consistency with entropy** *Let $\Omega$ be a continuous random variable, and $Q$ be a probability measure. If $U(\Omega)$ is the uniform distribution with the same support as $Q(\Omega)$, then we have*

$$\mathbb{D}\left[Q(\Omega) \,\|\, U(\Omega)\right] = \mathbb{H}\left[U(\Omega)\right] - \mathbb{H}\left[Q(\Omega)\right],$$

*where $\mathbb{H}$ denotes Shannon's entropy.*

**Additivity** *For any two discrete random variables $X, Y$ and probability measures $P, Q$*

$$\mathbb{D}\left[Q(X,Y) \,\|\, P(X,Y)\right] = \mathbb{D}\left[Q(X) \,\|\, P(X)\right] + \sum_x \mathbb{D}\left[Q(Y|x) \,\|\, P(Y|x)\right] Q(x).$$

*If $\mathbb{D}\left[\,\cdot\,\right]$ satisfies these properties, then it is the Kullback-Leibler divergence.*

*Proof.* Let $P(X)$ and $Q(X)$ be two arbitrary distributions for a discrete random variable X. We define a new continuous random variable $\Omega$ and let

$$p(\Omega = \omega \mid X = x) = \begin{cases} \frac{1}{P(x)} & 0 \le \omega \le P(X = x), \\ 0 & \text{otherwise}\cdot \end{cases}$$

Notice that both $P(\Omega|X)$ and $P(\Omega, X)$ are uniform distributions, and

$$\mathbb{H}\left[P(\Omega|x)\right] = \ln P(x),\ \mathbb{H}\left[P(\Omega, X)\right] = 0\cdot$$

We define $Q(\Omega|x)$ to be an arbitrary distribution that has the same support as $P(\Omega|x)$. This ensures that $Q(\Omega, X)$ and $P(\Omega, X)$ have the same support as well. Thus, we have

$$\begin{aligned} \mathbb{D}\left[Q(\Omega, X) \parallel P(\Omega, X)\right] &= \mathbb{H}\left[P(\Omega, X)\right] - \mathbb{H}\left[Q(\Omega, X)\right] \\ &= \sum_x \int q(\omega, x) \ln q(\omega, x) d\omega \\ &= \sum_x Q(x) \int q(\omega \mid x) \ln q(\omega, x) d\omega, \end{aligned}$$

and

$$\begin{aligned} \mathbb{D}\left[Q(\Omega|x) \parallel P(\Omega|x)\right] &= \mathbb{H}\left[P(\Omega|x)\right] - \mathbb{H}\left[Q(\Omega|x)\right] \\ &= \ln P(x) + \int q(\omega \mid x) \ln q(\omega \mid x) d\omega\cdot \end{aligned}$$

Now, we use the additivity property to obtain a formula for our divergence measure:

$$\begin{aligned} \mathbb{D}\left[Q(X) \parallel P(X)\right] &= \mathbb{D}\left[Q(\Omega, X) \parallel P(\Omega, X)\right] - \sum_x \mathbb{D}\left[Q(\Omega|x) \parallel P(\Omega|x)\right] Q(x) \\ &= \sum_x Q(x) \left( \int q(\omega \mid x) \ln \frac{q(\omega, x)}{q(\omega \mid x)} d\omega - \ln P(x) \right) \\ &= \sum_x Q(x) \left( \ln Q(x) \int q(\omega \mid x) d\omega - \ln P(x) \right) \\ &= \sum_x Q(x) \ln \frac{Q(x)}{P(x)}\cdot \end{aligned}$$

□

In this theorem, the first property simply states that the entropy of a distribution should be related to the divergence of the uniform distribution from it. The more the divergence is, the less the entropy must be.

The second property tells us that the divergence is an additive quantity. We know that the divergence of $P(X, Y)$ from $Q(X, Y)$ is a measure of the generic loss of using $P(X, Y)$ when we believe $Q(X, Y)$ is the true distribution. To calculate this loss, we can assume that $X$ happens before $Y$. In that way, this loss would include the loss of using $P(X)$, and then, if we observe $X = x$, which can happen with the probability $Q(x)$, it would also include the loss of using $P(Y \mid X = x)$. This property indicates that for combining these individual terms we should use addition.

We now briefly review some KL divergence properties needed in subsequent sections.

**Lemma 1.** *Let $X$ be a random variable, and let $P$, $Q$, and $R$ be probability distributions over $X$. The Kullback–Leibler divergence satisfies the following inequality:*

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right] \geq \sum_x Q(x) \ln \frac{R(x)}{P(x)}\cdot$$

*Proof.* The Kullback–Leibler divergence is nonnegative and is defined when $R$ is a probability distribution:

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, R(X)\right] \geq 0\cdot$$

By adding $\sum_x Q(x) \ln \frac{R(x)}{P(x)}$ to both sides of the above inequality, the desired inequality follows. □

**Theorem 2.** *Let $Q_1$, $Q_2$ and $P$ be probability distributions. If for some $0 \leq \alpha \leq 1$ we have $Q = \alpha Q_1 + (1 - \alpha) Q_2$, then:*

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right] \leq \alpha \mathbb{D}_{\mathrm{KL}}\left[Q_1(X) \,\|\, P(X)\right] + (1 - \alpha) \mathbb{D}_{\mathrm{KL}}\left[Q_2(X) \,\|\, P(X)\right]\cdot$$

*That is, the KL divergence is convex in its first argument.*

*Proof.* The result follows from Lemma 1 after expanding the left-hand side of the inequality using the definition of KL divergence. □

The KL divergence plays a key role in the notion of central distribution, which we introduce in the next section.

# 2 Central Distribution

Consider a random variable $X$. In probabilistic modeling, our knowledge about $X$ is typically represented by a probability distribution, which serves as a basis for making inferences.

However, in many situations, our information may be too limited to justify choosing a single distribution. Instead, we may only be able to specify a set of plausible distributions. This set can be viewed as a representation of *partial knowledge*, or equivalently, a state of lack of knowledge or *ignorance*.

A well-known example of this idea is the exchangeability assumption used in de Finetti's theorem. A sequence of random variables $X_1, X_2, \ldots$ is said to be exchangeable, if the ordering of the variables carries no information. That is, the joint probability distribution remains invariant under permutations of the variables. This assumption imposes a constraint, effectively narrowing the set of admissible distributions. Thus, exchangeability can be seen as a specific form of partial knowledge.

In this work, we adopt the perspective that a set of distributions provide a meaningful representation of the state of partial knowledge, and that this representation can likewise be used for inference.

Importantly, the act of choosing this set reflects some degree of knowledge. If even that knowledge is partial, we may similarly represent our uncertainty by a set of sets of distributions—capturing deeper layers of ignorance.

In Bayesian theory, any type of knowledge must ultimately be encoded in a single probability measure. Thus, to use our partial knowledge for performing inference in a Bayesian manner, we must find a principled way to convert our set-valued representation into a single probability measure.

The probability measure that we choose, represents our beliefs about the environment. It can be argued that this choice must depend solely on the environment. As long as the environment is the same, we must not change this choice. For example, if our loss function changes, or if the evidence that we get from the environment becomes more or less informative, none of these should change our beliefs about the environment, and therefore our chosen probability measure. We will use this principle several times in this discussion and refer to it as the *principle of objectivity*.

Subjectively, when we are uncertain about a set of distributions, a probability measure that has the lowest divergence from the distributions of the set can be considered the "safest" choice. "Safe" in the sense that using it incurs the least possible loss in the worst case, and more importantly, when convergence is possible for a set of distributions, if we use this probability measure, we are guaranteed to have a convergent distribution. By convergent distribution, we mean a distribution with sufficiently low divergence from any other distribution in that set.

**Definition 2.1** (Central Distribution)**.** Let $X$ be a random variable and $\mathcal{Q}$ be a set of probability distributions over $X$. We say a distribution $C$ is a *central distribution* of $\mathcal{Q}$ for $X$, if for every probability distribution $P$, we have

$$\sup_{Q\in\mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, C(X)\right] \leq \sup_{Q\in\mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right], \tag{2}$$

where $\mathbb{D}_{\mathrm{KL}}$ denotes the Kullback–Leibler (KL) divergence.

Note that, under this definition, the central distribution of $\mathcal{Q}$ is not necessarily an element of $\mathcal{Q}$. That is, the distribution that best represents the set in the specified sense may lie outside the set it summarizes.

**Definition 2.2** (Interior Distribution)**.** Let $\mathcal{Q}$ be a set of probability distributions over a random variable $X$. A distribution $R$ is called an *interior distribution* of $\mathcal{Q}$ if, for every probability distribution $P$, there exists some $Q \in \mathcal{Q}$ such that

$$\mathbb{D}_{\mathrm{KL}}\left[R(X) \,\|\, P(X)\right] \leq \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right],$$

**Theorem 3.** *Any distribution that can be expressed as a convex combination of elements of $\mathcal{Q}$ is an interior distribution of $\mathcal{Q}$.*

*Proof.* Let $R = \sum_i \alpha_i Q_i$, where $Q_i \in \mathcal{Q}$, $\alpha_i \geq 0$, and $\sum_i \alpha_i = 1$.

Fix an arbitrary probability distribution $P$. By the convexity of KL divergence in its first argument, we have:

$$\mathbb{D}_{\mathrm{KL}}\left[R(X) \,\|\, P(X)\right] \leq \sum_i \alpha_i \mathbb{D}_{\mathrm{KL}}\left[Q_i(X) \,\|\, P(X)\right] \cdot$$

Now, note that the right-hand side is a weighted average of the KL divergences. So, there must be at least one index $j$ such that

$$\mathbb{D}_{\mathrm{KL}}\left[R(X) \,\|\, P(X)\right] \leq \mathbb{D}_{\mathrm{KL}}\left[Q_j(X) \,\|\, P(X)\right] \cdot$$

Since $Q_j \in \mathcal{Q}$, the condition in Definition 2.2 is satisfied, and $R$ is indeed an interior distribution of $\mathcal{Q}$. □

**Theorem 4.** *Let $R$ be an interior distribution of $\mathcal{Q}$. If $C$ is a central distribution of $\mathcal{Q}$, it is also a central distribution of $\mathcal{Q} \cup \{R\}$ and vice versa.*

*Proof.* This theorem follows directly from the definition of an interior distribution, which implies that for any probability distribution $P$, we have

$$\sup_{Q\in\mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right] = \sup_{Q\in\mathcal{Q}\cup\{R\}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right] \cdot$$ □

One important consequence of this theorem is that, in order to identify a central distribution within a parametric model, it suffices to restrict attention to the set of likelihood distributions induced by different values of the model parameters.

The next theorem is a standard result that is typically used in the derivation of central distributions.

**Theorem 5.** *Let $X$ and $T$ be two random variables, and let $\mathcal{Q}$ be a set of distributions over the pair $(X, T)$. Assume that for every $Q \in \mathcal{Q}$, we have*

$$Q(X, T) = R(X \mid T)Q(T),$$

*where $R$ is a fixed conditional distribution. If $C$ is a central distribution of $\mathcal{Q}$ for $T$, then $R(X \mid T)C(T)$ is a central distribution of $\mathcal{Q}$ for $(X, T)$.*

*Proof.* Let $C$ be a central distribution of $\mathcal{Q}$ for $T$. For an arbitrary probability distribution $P$ we have

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}} \left[ Q(X,T) \,\|\, R(X \mid T)C(T) \right] \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}} \left[ Q(X,T) \,\|\, R(X \mid T)P(T) \right] \cdot \tag{3}$$

Since the KL divergence is nonnegative, for every probability distribution $P$ and every $Q \in \mathcal{Q}$, we have

$$\mathbb{D}_{\mathrm{KL}} \left[ Q(X,T) \,\|\, R(X \mid T)P(T) \right] \leq \mathbb{D}_{\mathrm{KL}} \left[ Q(X,T) \,\|\, P(X,T) \right],$$

which implies

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}} \left[ Q(X,T) \,\|\, R(X \mid T)P(T) \right] \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}} \left[ Q(X,T) \,\|\, P(X,T) \right] \cdot$$

Therefore, combining this inequality with Inequality 3, we conclude that $R(X \mid T)C(T)$ is a central distribution of $\mathcal{Q}$ for $(X, T)$. $\square$

The central distribution is essentially the solution to a convex-concave game. Since the solution to a convex-concave game corresponds to a saddle point, by defining the appropriate objective function, we can characterize the central distribution using standard convex optimization techniques.

**Definition 2.3** (Saddle Point)**.** Consider a function $f : \mathcal{W} \times \mathcal{Z} \to \mathbb{R}$. We refer to a pair $(w^*, z^*)$ as a *saddle point* for $f$, if we have

$$f(w^*, z) \leq f(w^*, z^*) \leq f(w, z^*),$$

for all $w \in \mathcal{W}$ and $z \in \mathcal{Z}$.

Next, we will establish the connection between the saddle point and the central distribution. One significant insight provided by the next theorem is that it suggests inference and searching for a central distribution can be combined and performed simultaneously when finding a saddle point.

**Theorem 6.** *Let $X$ be a discrete random variable, and $\mathcal{Q}$ be a finite set of distributions over $X$. Consider the following function:*

$$f(P,R) = \sum_i \mathbb{D}_{\mathrm{KL}}\left[Q_i(X) \parallel P(X)\right] R(i),$$

*where $Q_i \in \mathcal{Q}$, and $P$, $R$ are probability distributions.*

*If $(P^*, R^*)$ is a saddle point of $f$, then $P^*$ is a central distribution of $\mathcal{Q}$ for $X$.*

*Proof.* Since $X$ is a discrete random variable, $\mathcal{Q}$ has a central distribution for $X$. Let $C$ be a central distribution of $\mathcal{Q}$ for $X$, and $(P^*, R^*)$ be a saddle point of $f$. As $C$ is a probability distribution, the point $(C, R^*)$ lies within the domain of $f$. By the definition of a saddle point, this implies that

$$f(P^*, R^*) \leq f(C, R^*) \cdot \tag{4}$$

Because $R^*$ is a probability distribution, the definition of $f$ implies:

$$f(C, R^*) \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \parallel C(X)\right] \cdot$$

On the other hand,

$$f(P^*, R^*) = \max_R f(P^*, R) = \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \parallel P^*(X)\right] \cdot$$

Thus, Inequality 4 implies

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \parallel P^*(X)\right] \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \parallel C(X)\right] \cdot$$

Since $C$ is a central distribution, we must have

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \parallel P^*(X)\right] = \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(X) \parallel C(X)\right],$$

and $P^*$ is also a central distribution of $\mathcal{Q}$ for $X$. $\square$

**Lemma 2.** *Let $f$ be the function defined in theorem 6. The point $(P^*, R^*)$ is a saddle point for $f$, if and only if there are constants $\lambda$, $\eta$ and a function $\mu$, such that for every $i$, $x$ the following conditions hold:*

$$\sum_x Q_i(x) \ln \frac{Q_i(x)}{P^*(x)} - \mu(i) = \lambda, \tag{5}$$

$$P^*(x) = \sum_i R^*(i) Q_i(x) \tag{6}$$

$$\sum_i R^*(i) = 1, \tag{7}$$

$$R^*(i) \geq 0, \tag{8}$$

$$\mu(i) \geq 0,$$

$$\mu(i) R^*(i) = 0 \cdot \tag{9}$$

*Furthermore, for a fixed probability distribution $R$, a distribution $\hat{P}$ minimizes $f(\,\cdot\,, R)$, if and only if for all $x$ it satisfies*

$$\hat{P}(x) = \sum_i Q_i(x) R(i) \cdot \tag{10}$$

*Proof.* Similar to $f$, we define $\hat{f}$ as follows:

$$\hat{f}(P, R) = \sum_i R(i) \sum_x Q_i(x) \ln \frac{Q_i(x)}{P(x)},$$

where $P$, $R$ are arbitrary real functions instead of probability distributions. This enables us to formulate the saddle point of $f$ as the solution to a constrained optimization problem.

Let $(P^*, R^*)$ be a simultaneous solution to the following two optimization problems:

$$\begin{array}{lll} \text{minimize} & \hat{f}(P, R) & \text{w.r.t. } P \\ \text{subject to} & \sum_x P(x) = 1, & \\ & & \\ \text{maximize} & \hat{f}(P, R) & \text{w.r.t. } R \\ \text{subject to} & R(i) \geq 0, & \text{for all } i \\ & \sum_i R(i) = 1 & \end{array} \tag{11}$$

Clearly, if $(P^*, R^*)$ exists, it will be a saddle point for $f$.

Both of these optimization problems are convex [4] and differentiable. To establish this, consider that all the constraint functions are affine, and therefore are convex and differentiable.

Additionally, $\hat{f}$ is differentiable function with respect to both $R$ and $P$. For any fixed $P$, $\hat{f}(P,\cdot)$ is an affine function and is concave.

On the other hand, for a fixed solution to the second optimization problem denoted by $\hat{R}$, we can write

$$\hat{f}(P,\hat{R}) = c - \sum_{i,x} \hat{R}(i)Q_i(x) \ln P(x), \tag{12}$$

where $c$ is not a function of $P$. Since $\hat{R}$ is a solution to the second optimization problem, it satisfies the corresponding constraints and is therefore a valid probability distribution. Hence, for all $i$, $x$ we have $\hat{R}(i)Q_i(x) \geq 0$, and $\hat{f}(P,\cdot)$ is the negative of a linear combination of concave $\ln P(x)$ functions with nonnegative coefficients. This function is convex.

Thus, both optimization problems are convex and differentiable, with affine constraints. The constraints simply restrict the feasible set to the set of probability distributions, and it is always possible to find feasible points that meet all the constraints. Therefore, Slater's condition is satisfied, and the KKT conditions provide necessary and sufficient conditions for optimality [4]. In other words, If a point $(P^*, R^*)$ satisfies the KKT conditions for both problems, it will be a saddle point for $f$, and vice versa.

The KKT conditions are as follows for every $x$, $i$ and constants $\lambda$, $\eta$:

$$\sum_i R^*(i)Q_i(x) = \eta P^*(x) \tag{13}$$

$$\sum_x P^*(x) = 1, \tag{14}$$

$$\begin{gathered} \sum_x Q_i(x) \ln \frac{Q_i(x)}{P^*(x)} - \mu(i) = \lambda, \\ \sum_i R^*(i) = 1, \\ R^*(i) \geq 0, \\ \mu(i) \geq 0, \\ \mu(i)R^*(i) = 0\cdot \end{gathered} \tag{15}$$

By summing Equation 13 over $x$ and using Equation 15, we have

$$\eta = 1\cdot$$

Thus, we get Equation 6 which also implicitly implies Equation 14.

To prove the second part of the lemma, we observe that Equations 13 and 14 are the KKT conditions for the minimization problem in 11, when $R$ is a probability distribution. Therefore, these conditions provide necessary and sufficient conditions for a point to be a minimizer of $f(\,\cdot\,, R)$, where $R$ is a fixed distribution. $\square$

**Lemma 3.** *Let $\mathcal{Q}$ be a finite set of probability distributions over a discrete random variable $X$. Define the function $g$ as:*

$$g(R) = \sum_i \mathbb{D}_{\mathrm{KL}}\left[Q_i(X) \,\|\, R(X)\right] R(i),$$

*where $Q_i \in \mathcal{Q}$, and $R$ is a probability distribution such that*

$$R(X, I) = Q_i(X) R(I)\cdot$$

*Let $f$ be the function defined in theorem 6. If $(P^*, R^*)$ is a saddle point for $f$, then the joint distribution $Q_i(X)R^*(I)$ is a maximizer of $g$.*

*Proof.* Let $(P^*, R^*)$ be a saddle point of $f$. By Lemma 2, $P^*$ satisfies Equation 6, and thus:

$$g(R^*) = f(P^*, R^*)\cdot$$

Now, let $R$ be an arbitrary probability distribution. From Lemma 2, it follows that $g(R)$ indeed corresponds to the minimum of $f(\,\cdot\,, R)$, which implies

$$g(R) \leq f(P^*, R)\cdot$$

On the other hand, since $(P^*, R^*)$ is a saddle point, we also have

$$f(P^*, R) \leq f(P^*, R^*)\cdot$$

Combining the two inequalities, we get:

$$g(R) \leq g(R^*)\cdot$$

Since $R$ was an arbitrary distribution, this shows that $g(R^*)$ is the maximum of $g$, as desired. $\square$

The following theorem provides the rationale behind the term "central distribution." Exactly like the center of a circle, the central distribution is equidistant from the boundary.

**Theorem 7** (Centrality)**.** *Let $X$ be a discrete random variable, and let $\mathcal{Q}$ be a finite set of distributions over $X$. Suppose $C$ is a convex combination of elements of $\mathcal{Q}$, with all weights strictly positive. If for some constant $\lambda$ and every $Q \in \mathcal{Q}$ we have:*

$$\mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, C(X)\right] = \lambda,$$

*then $C$ is a central distribution of $\mathcal{Q}$ for $X$.*

*Proof.* Lemma 2 provides a set of sufficient conditions for a saddle point. Note that more restrictive conditions can also be imposed to derive an alternative set of sufficient conditions. We can replace Inequality 8 with the more restrictive $R^*(i) > 0$ condition. By Equation 9, that implies $\mu(i) = 0$, and Lemma 2 conditions can be simplified to

$$\sum_x Q_i(x) \ln \frac{Q_i(x)}{P^*(x)} = \lambda,$$
$$P^*(x) = \sum_i R^*(i) Q_i(x),$$

which must hold for every $Q_i \in \mathcal{Q}$ and some strictly positive probability distribution $R^*$.

Since $C$ satisfies these condition, Lemma 2 and Theorem 6 imply that $C$ is a central distribution of $\mathcal{Q}$ for $X$. □

**Theorem 8** (Imperfection)**.** *Assuming the conditions of Theorem 7 and using the same $\lambda$, if $P$ is a convex combination of elements of $\mathcal{Q}$, then there exists some $Q_0 \in \mathcal{Q}$ such that*

$$\mathbb{D}_{\mathrm{KL}}\left[Q_0(X) \,\|\, P(X)\right] \leq \lambda\cdot$$

*Proof.* We proceed by contradiction. Suppose that for every $Q \in \mathcal{Q}$, we have $\mathbb{D}_{\mathrm{KL}}\left[Q(X) \,\|\, P(X)\right] > \lambda$. This implies that

$$g(P) > \lambda = g(C),$$

where $C$ is the distribution specified in Theorem 7, and $g$ is the function defined in Lemma 3. Note that $g(P)$ is defined, because $P$ is a convex combination of elements of $\mathcal{Q}$.

However, by the conditions of Theorem 7, the distribution $C$ corresponds to a saddle point of $f$. This leads to a contradiction, since Lemma 3 asserts that a saddle point of $f$ must be a maximizer of $g$. Therefore, our assumption must be false. □

These theorems were proved under the assumption that $\mathcal{Q}$ is finite; however, they can be extended to cases where $\mathcal{Q}$ can be indexed by a finite-dimensional real vector $\Theta$. We can treat this vector as a random variable and interpret $Q_\theta$ as a conditional distribution $Q(x \mid \theta)$. Then, the weight function, $R(i)$, will essentially become a prior distribution over $\theta$.

The next theorem shows that there is a strong relationship between central distributions and Bernardo's reference priors [3].

**Theorem 9.** *Let $X$ be a discrete random variable, and let $\mathcal{Q}$ be a set of probability distributions over $X$, indexed by a finite-dimensional random vector $\Theta$. Let $c$ be a strictly positive prior distribution for $\Theta$ such that for $Q(x \mid \theta) \in \mathcal{Q}$ we have*

$$c(\theta) \propto \exp\left\{\sum_x Q(x \mid \theta) \ln c(\theta \mid x)\right\}\cdot \tag{16}$$

*Then, $C(x) = \int Q(x \mid \theta)c(\theta)d\theta$ is a central distribution of $\mathcal{Q}$ for $X$. We also refer to $c$ as a central prior for $X$.*

*Proof.* From Equation 16, it follows that there exists a constant $\lambda$ such that

$$c(\theta) = \exp\left\{-\lambda + \sum_x Q(x \mid \theta) \ln \frac{Q(x \mid \theta)c(\theta)}{C(x)}\right\}\cdot$$

Since $c(\theta)$ is strictly positive, we have

$$\begin{aligned} c(\theta) &= \exp\left\{-\lambda + \mathbb{D}_{\mathrm{KL}}\left[Q_\theta(X) \,\|\, C(X)\right] + \ln c(\theta)\right\} \\ &= c(\theta) \exp\left\{-\lambda + \mathbb{D}_{\mathrm{KL}}\left[Q_\theta(X) \,\|\, C(X)\right]\right\}, \end{aligned}$$

which implies

$$\mathbb{D}_{\mathrm{KL}}\left[Q_\theta(X) \,\|\, C(X)\right] = \lambda\cdot$$

Since this holds for all $\theta$, and $c$ is strictly positive, by Theorem 7 it follows that $C$ is a central distribution of $\mathcal{Q}$ for $X$. $\square$

Intuitively, a central distribution acts as a representative or summary of a set of distributions. While it may not belong to the set itself, it reflects the overall behavior of the set in a way that is meaningful for inference and also for evaluating other distributions. In other words, it provides a meaningful way for comparing distributions based on our partial knowledge.

**Definition 2.4** (Central Divergence)**.** For any distribution $P$, we define the *central divergence* of $P$ from a set of distributions $\mathcal{Q}$, with respect to a random variable $X$, as follows:

$$\mathbb{D}_{\mathrm{CD}}\left[\mathcal{Q} \,\|\, P(X)\right] = \inf_C \mathbb{D}_{\mathrm{KL}}\left[C(X) \,\|\, P(X)\right],$$

where $C$ denotes a central distribution of $\mathcal{Q}$ for $X$.

Note that when $\mathcal{Q}$ has a unique central distribution, the central divergence is simply defined as the KL divergence from that unique central distribution.

Sometimes, partial knowledge is represented by multiple sets of distributions. Our current definition of the central distribution does not account for this case. The following definition extends the concept to handle such situations. While not directly about feature selection, this extension plays a key role in enabling the connection between central distributions and feature selection, a connection that will become clearer later.

**Definition 2.5.** Let $X$ be a random variable, and let $\mathcal{Q}^*$ be a set whose elements are sets of distributions over $X$. Let $C_{\mathcal{Q}}$ be a central distribution of $\mathcal{Q} \in \mathcal{Q}^*$ for $X$. We say $C^*$ is a central distribution of $\mathcal{Q}^*$, if for every probability distribution $P$ we have:

$$\sup_{\mathcal{Q}\in\mathcal{Q}^*} \mathbb{D}_{\mathrm{CD}}\left[\mathcal{Q} \,\|\, C^*(X)\right] \leq \sup_{\mathcal{Q}\in\mathcal{Q}^*} \mathbb{D}_{\mathrm{CD}}\left[\mathcal{Q} \,\|\, P(X)\right],$$

where $\mathbb{D}_{\mathrm{CD}}\left[\mathcal{Q} \,\|\, \cdot\right]$ refers to the central divergence as defined in Definition 2.4.

Suppose we want to model an exchangeable sequence of random variables. Let $S_n$ denote the sequence of the first $n$ variables: $S_n = (X_1, X_2, \ldots, X_n)$. As discussed earlier, exchangeability can be interpreted as a form of partial knowledge, which allows us to represent our uncertainty using a central distribution over $S_n$.

However, this central distribution generally depends on the sequence length $n$. According to the principle of objectivity, our beliefs should not depend on the specifics of our experimental setup, and in particular, should not vary with the length of the sequence.

It is important to note that a distribution defined over a longer sequence induces marginal distributions over its shorter subsequences. This observation suggests that, in order to satisfy the principle of objectivity, we may instead determine a central distribution over an infinite exchangeable sequence, from which the distributions of all finite initial segments can be derived.

This requires constructing a central distribution that is well-defined over $S_\infty$.

**Definition 2.6.** Let $\{S_n\}_{n=1}^{\infty}$ be a sequence of random variables and $\mathcal{Q}$ be a set of probability measures, where each element of $\mathcal{Q}$ defines a distribution over each $S_n$. Assume $C_n$ is a central distribution for $S_n$. Let $C$ be a probability measure that defines a distribution over every $S_n$. $C$ is a central distribution of $\mathcal{Q}$ for $\{S_n\}_{n=1}^{\infty}$, if we have

$$\lim_{n\to\infty} \mathbb{D}_{\mathrm{KL}}\left[C_n(S_n) \,\|\, C(S_n)\right] = 0\cdot$$

Intuitively, this definition states that as $n \to \infty$ the distribution that $C$ induces over $S_n$ becomes increasingly similar to a central distribution for $S_n$. In other words, $C$ is a central distribution for $S_\infty$.

Our next goal is to extend Theorem 5 so that it also applies in this asymptotic setting, thereby providing a characterization of central distributions for infinite sequences.

**Theorem 10.** *Let $\{S_n\}_{n=1}^\infty$ and $\{T_n\}_{n=1}^\infty$ be two sequences of random variables. Let $\mathcal{Q}$ be a set of probability measures, where each $Q \in \mathcal{Q}$ defines a joint distribution over $(S_n, T_n)$. Assume there exists a natural number $M$ and a fixed conditional distribution $R$ such that, for all $Q \in \mathcal{Q}$ and $n \geq M$,*

$$Q(S_n, T_n) = R(S_n \mid T_n)Q(T_n).$$

*Let $C$ be a central distribution of $\mathcal{Q}$ for $\{T_n\}_{n=1}^\infty$ (Definition 2.6). Define a probability measure $C^*$ by*

$$C^*(S_n, T_n) = R(S_n \mid T_n)C(T_n), \quad n \geq M.$$

*Then $C^*$ is a central distribution for $\{(S_n, T_n)\}_{n=1}^\infty$.*

*Proof.* By Theorem 5, for each $n \geq M$ we have

$$C_n(S_n, T_n) = R(S_n \mid T_n)C_n(T_n),$$

where $C_n(T_n)$ is a central distribution for $T_n$, and $C_n(S_n, T_n)$ is a central distribution for $(S_n, T_n)$. Therefore,

$$\begin{aligned} \mathbb{D}_{\mathrm{KL}}\left[C_n(S_n, T_n) \,\|\, C^*(S_n, T_n)\right] &= \mathbb{D}_{\mathrm{KL}}\left[R(S_n \mid T_n)C_n(T_n) \,\|\, R(S_n \mid T_n)C(T_n)\right] \\ &= \mathbb{D}_{\mathrm{KL}}\left[C_n(T_n) \,\|\, C(T_n)\right]. \end{aligned}$$

Since $C$ is a central distribution for $\{T_n\}_{n=1}^\infty$ we have

$$\lim_{n\to\infty} \mathbb{D}_{\mathrm{KL}}\left[C_n(T_n) \,\|\, C(T_n)\right] = 0.$$

It follows that

$$\lim_{n\to\infty} \mathbb{D}_{\mathrm{KL}}\left[C_n(S_n, T_n) \,\|\, C^*(S_n, T_n)\right] = 0,$$

so $C^*$ is a central distribution for $\{(S_n, T_n)\}_{n=1}^\infty$. $\square$

We now present an asymptotic version of Theorem 7, applicable to infinite sequences of random variables.

**Theorem 11.** *Let $\{S_n\}_{n=1}^{\infty}$ be a sequence of random variables and $\{\lambda_n\}_{n=1}^{\infty}$ be a sequence of constants. If $C$ is a weighted average of elements of $\mathcal{Q}$ with strictly positive weights such that for every $Q \in \mathcal{Q}$ we have*

$$\lim_{n\to\infty} \left(\mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \lambda_n\right) = 0,$$

*then $C$ is a central distribution of $\mathcal{Q}$ for $\{S_n\}_{n=1}^{\infty}$.*

*Proof.* For the purpose of this proof, we additionally assume—perhaps unnecessarily—that for each $S_n$, there exists a central distribution $C_n$ that satisfies the conditions of Theorem 7. That is, there exists a constant $\mu_n$ such that for every $Q \in \mathcal{Q}$

$$\mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C_n(S_n)\right] = \mu_n\cdot$$

For each $n$, by the definition of central distribution, we have

$$\sup_{Q\in\mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] \geq \mu_n\cdot$$

Moreover, by Theorem 8, it follows that

$$\inf_{Q\in\mathcal{Q}} \mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] \leq \mu_n\cdot$$

Now, by the squeeze theorem, the sequence $\{\mu_n - \lambda_n\}_{n=1}^{\infty}$ must converge to 0. Therefore, for any arbitrary $Q \in \mathcal{Q}$, we have

$$\begin{aligned}
\lim_{n\to\infty} \sum_{s_n} Q(s_n) \ln \frac{C_n(s_n)}{C(s_n)} &= \lim_{n\to\infty} \left(\mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C_n(S_n)\right]\right) \\
&= \lim_{n\to\infty} \left(\mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \lambda_n - \mu_n + \lambda_n\right) \\
&= \lim_{n\to\infty} \left(\mathbb{D}_{\mathrm{KL}}\left[Q(S_n) \,\|\, C(S_n)\right] - \lambda_n\right) - \lim_{n\to\infty} (\mu_n - \lambda_n) \\
&= 0\cdot
\end{aligned}$$

Since $\mathbb{D}_{\mathrm{KL}}\left[C_n(S_n) \,\|\, C(S_n)\right]$ is a weighted average of $\sum_{s_n} Q(s_n) \ln \frac{C_n(s_n)}{C(s_n)}$, we can conclude that

$$\lim_{n\to\infty} \mathbb{D}_{\mathrm{KL}}\left[C_n(S_n) \,\|\, C(S_n)\right] = 0\cdot$$

Therefore, $C$ is a central distribution of $\mathcal{Q}$ for the sequence $\{S_n\}_{n=1}^{\infty}$. □

If we have a consistent estimator sequence $\{T_n\}_{n=1}^{\infty}$ for the index of a set, the following theorem provides a way to construct a central distribution for that sequence. Here, by a consistent estimator we mean a sequence of estimators such that, for any fixed index value, the distribution of $T_n$ becomes increasingly concentrated around the corresponding value of the index as $n$ increases. A similar result can also be found in [3].

**Theorem 12.** *Let $\{T_n\}_{n=1}^{\infty}$ be a sequence of random variables, and let $\mathcal{Q}$ be a set of probability measures, where each element of $\mathcal{Q}$ defines a distribution over each $T_n$. Assume that $\mathcal{Q}$ can be indexed by a finite-dimensional vector $\Theta$. If $\{T_n\}_{n=1}^{\infty}$ is a consistent sequence of estimators for $\Theta$, and there exists a sequence of constants $\{\mu_n\}_{n=1}^{\infty}$ such that the sequence $\left\{\frac{P(\theta|T_n=\theta)}{\mu_n}\right\}_{n=1}^{\infty}$ converges for every value of $\theta$, then*

$$c(\theta) \propto \lim_{n\to\infty} \frac{P(\theta \mid T_n = \theta)}{\mu_n}, \tag{17}$$

*is a central prior of $\mathcal{Q}$ for $\{T_n\}_{n=1}^{\infty}$.*

*Proof.* Since $T_n$ is a consistent estimator, the sampling distribution $Q\left(T_n \mid \theta\right)$ concentrates around $T_n = \theta$. Moreover, since ln is a continuous function, it follows from Equation 17 that

$$\lim_{n\to\infty} \left( \sum_{t_n} Q(t_n \mid \theta) \ln P(\theta \mid t_n) - \ln \mu_n \right) + \ln k = \ln c(\theta),$$

where $k$ is a normalizing constant converting proportionality into equality. As the posterior distribution of a consistent estimator is asymptotically independent of the prior, we may replace the prior with $c(\theta)$:

$$\lim_{n\to\infty} \left( \sum_{t_n} Q(t_n \mid \theta) \ln \frac{c(\theta \mid t_n)}{c(\theta)} - \ln \mu_n + \ln k \right) = 0\cdot$$

Applying Bayes' rule and letting $\lambda_n = \ln \mu_n - \ln k$, for every value of $\theta$ we obtain:

$$\lim_{n\to\infty} \left( \mathbb{D}_{\mathrm{KL}} \left[ Q\left(T_n \mid \theta\right) \| C(T_n) \right] - \lambda_n \right) = 0\cdot$$

Thus, by Theorem 11, we can conclude that $c(\theta)$ is a central prior of $\mathcal{Q}$ for $\{T_n\}_{n=1}^{\infty}$. □

A central distribution over an estimator cannot, by itself, be used to perform inference at the level of observed outcomes. To do so, we must convert it into a distribution over the primary variables of interest. Theorems 5 and 10 provide a simple method for this conversion.

**Example 2.1** (Binomial Experiment)**.** Consider a binomial experiment with $n$ Bernoulli trials. Denote the sequence of outcomes by $S_n$, and let $k$ be the number of successes observed in $S_n$

Let $\mathcal{Q}$ be the set of distributions corresponding to different values of the success probability $\theta \in [0, 1]$. We can consider $\theta$ as an index parameter for

$\mathcal{Q}$, and we write $Q(S_n \mid \theta)$ to denote the distribution $Q_\theta$ applied to $S_n$. Explicitly,

$$Q(S_n \mid \theta) = \theta^k (1-\theta)^{n-k}.$$

Define $T_n$ to be the proportion of successes in $n$ trials. Note that $T_n$ is a consistent estimator for $\theta$. Therefore, we can use Theorem 12 to find a central distribution for the sequence $\{T_n\}_{n=1}^{\infty}$.

Assuming a uniform prior on $\theta$, Bayes' rule gives the posterior distribution

$$P\left(\theta \,\middle|\, T_n = \tfrac{k}{n}\right) = (n+1)\binom{n}{k}\theta^k(1-\theta)^{n-k},$$

which is a Beta$(k+1, n-k+1)$ distribution. Using Stirling's formula, we derive its asymptotic form:

$$P\left(\theta \,\middle|\, T_n = \tfrac{k}{n}\right) \sim \frac{(n+1)n^{(n+\frac{1}{2})}}{\sqrt{2\pi}(n-k)^{(n-k+\frac{1}{2})}k^{(k+\frac{1}{2})}}\,\theta^k(1-\theta)^{n-k}.$$

We now let $T_n \to \theta$:

$$P\left(\theta \mid T_n = \theta\right) \sim \frac{n+1}{\sqrt{2\pi n}}\,\theta^{-\frac{1}{2}}(1-\theta)^{-\frac{1}{2}}.$$

Therefore, if we define

$$\mu_n = \frac{n+1}{\sqrt{2\pi n}},$$

the following limit exists:

$$\lim_{n\to\infty} \frac{P\left(\theta \mid T_n = \theta\right)}{\mu_n} = \theta^{-\frac{1}{2}}(1-\theta)^{-\frac{1}{2}}.$$

By Theorem 12, this implies that $c(\theta) \propto \theta^{-\frac{1}{2}}(1-\theta)^{-\frac{1}{2}}$ is a central prior for $\{T_n\}_{n=1}^{\infty}$.

Now we apply Theorem 10 to obtain a central distribution for $\{S_n\}_{n=1}^{\infty}$. Since $T_n$ is a deterministic function of $S_n$, the joint distribution $Q(S_n, T_n)$ coincides with the distribution of $S_n$: it assigns probability $Q(S_n)$ when $T_n$ is compatible with $S_n$, and zero otherwise. In other words, knowledge of $(S_n, T_n)$ is equivalent to knowledge of $S_n$.

In the binomial experiment, all sequences with the same number of successes and failures have equal probability. Therefore,

$$Q(S_n \mid \theta) = Q\big(S_n,\, T_n = \tfrac{k}{n} \mid \theta\big) = \frac{1}{\binom{n}{k}} Q(T_n \mid \theta).$$

Let $C^*$ denote the central distribution of $\{S_n\}_{n=1}^{\infty}$. By Theorem 10, the distribution induced by $C^*$ on each $S_n$ is given by:

$$C^*(S_n) = \frac{\Gamma\left(k+\frac{1}{2}\right)\Gamma\left(n-k+\frac{1}{2}\right)}{\Gamma\left(n+1\right)\Gamma\left(\frac{1}{2}\right)^2}.$$

When dealing with layered ignorance and attempting to determine the central distribution of a set of sets of distributions, as defined in Definition 2.5, we typically consider the case where the number of such sets is finite.

In this setting, and under appropriate conditions, the central distribution reduces to the uniform distribution.

**Corollary 12.1.** *Under the assumptions of Theorem 12, if $\mathcal{Q}$ is a countable set indexed by a discrete random variable $I$, then $C(i) \propto 1$ is a central prior of $\mathcal{Q}$ for $\{T_n\}_{n=1}^{\infty}$.*

We now present some examples of central distributions and demonstrate how they can be used for feature selection.

**Example 2.2** (Fused Multinomial Model)**.** Let $\{S_n\}_{n=1}^{\infty}$ be a sequence where each $S_n = (X_1, X_2, \ldots, X_n)$ represents $n$ independent trials of an experiment with $k$ possible outcomes, and $\mathcal{Q}$ be a set of probability distributions over $S_n$ indexed by a real vector $\theta$. Let $f : \mathcal{X} \to \mathcal{Y}$ be a deterministic function that maps each outcome of $X_i$ to a label $Y_i = f(X_i)$, where $\mathcal{Y} = \{1, \ldots, m\}$. Assume that for all $i$ and $Q_\theta \in \mathcal{Q}$, if $f(x_i) = f(x_i')$ we have

$$Q_\theta(x_1, x_2, \ldots, x_n) = Q_\theta(x_1', x_2', \ldots, x_n').$$

That is, the conditional distribution $Q_\theta(X_i \mid Y_i = y)$ is uniform over the preimage $f^{-1}(y)$.

We can consider $\theta = (\theta_1, \ldots, \theta_m)$ to be a probability vector over $\mathcal{Y}$, where each $\theta_y$ specifies the probability of observing label $y$. For a sequence $S_n$, let $n_y$ denote the number of times label $y \in \mathcal{Y}$ appears. Let $|f^{-1}(y)|$ denote the number of outcomes in $\mathcal{X}$ that are mapped to label $y$ by $f$. Then the probability of observing $S_n$ under $Q_\theta \in \mathcal{Q}$, denoted $Q(S_n \mid \theta)$, is:

$$Q(S_n \mid \theta) = \prod_{y=1}^{m} \left( \frac{1}{|f^{-1}(y)|} \cdot \theta_y \right)^{n_y}.$$

Now, consider the estimator sequence $T_n = \frac{1}{n}(n_1, n_2, \ldots, n_m)$, which is a consistent estimator for $\theta$. The conditional distribution $Q(S_n \mid T_n)$ is then given by:

$$Q(S_n \mid T_n) = \frac{1}{n!} \prod_{y=1}^{m} \frac{n_y!}{|f^{-1}(y)|^{n_y}}.$$

Note that this distribution is independent of $Q_\theta$. Therefore, by Theorem 5, a central distribution for $S_n$ can be obtained from a central distribution for $T_n$.

As $n \to \infty$, the distribution of $T_n$ converges to a multivariate Gaussian distribution. Using this fact and Theorem 12, we can determine a central prior for $\{T_n\}_{n=1}^\infty$:

$$c(\theta) \propto \prod_{y=1}^{m} \theta_y^{-\frac{1}{2}}$$

This is a Dirichlet prior and induces a Dirichlet-multinomial distribution for each $T_n$:

$$C(T_n) = \frac{\Gamma\left(\frac{m}{2}\right)\Gamma\left(n+1\right)}{\Gamma\left(n+\frac{m}{2}\right)} \prod_{y=1}^{m} \frac{\Gamma\left(n_y+\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}\right)\Gamma\left(n_y+1\right)}.$$

Note that here, $C$ denotes a central distribution for $T_\infty$, and $C(T_n)$ refers to the marginal distribution induced on $T_n$.

By Theorem 5, we can derive a central distribution for $\{S_n\}_{n=1}^\infty$, which induces the following distribution over each $S_n$:

$$C_f(S_n) = \frac{\Gamma\left(\frac{m}{2}\right)}{\Gamma\left(n+\frac{m}{2}\right)} \prod_{y=1}^{m} \frac{\Gamma\left(n_y+\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}\right)|f^{-1}(y)|^{n_y}}.$$

# 3 Illustrative Examples

## 3.1 Attribute Selection

In this example, we consider a classification problem where the goal is to predict a target class, denoted by $Y$, from a set of observed nominal attributes $(A_1, A_2, \ldots, A_n)$. We assume that $Y$ may depend on an unknown subset of the attributes, rather than the full set.

This reflects a form of sparsity in the relationship between the class and the attributes, where irrelevant features are ignored in the true generative model.

We can use the fused multinomial model of Example 2.2 to capture this relationship. As an example, assume that $Y$ depends only on $A_2$, $A_5$. Then, we define the labeling function $f$ as follows:

$$f(A_1, A_2, \ldots, A_n, Y) = (A_2, A_5, Y)\cdot$$

In a fused multinomial model with this labeling function, $Y$ will be conditionally independent from other attributes, given $A_2, A_5$. Additionally, $P(A_1, A_2, \ldots, A_n \mid A_2, A_5)$ will be uniform.

For every possible subset of attributes a distinct fused multinomial model of this form is needed. Each model corresponds to a set of plausible distributions. Definition 2.5 provides a way to define a central distribution for situations like this one.

By Theorem 7 we know that this central distribution would be a weighted average of the central distributions of individual fused multinomial models. Since we have a finite number of models, Corollary 12.1 implies that these weights are uniform:

$$C_{\mathcal{Q}^*}(S_n) \propto \sum_i C_{f_i}(S_n)\cdot$$

To establish this, observe that we can construct a consistent estimator for the model indicator $I$ by checking for the equality of different outcomes. Specifically, the estimator identifies outcomes with equal frequency in the data and returns the indicator for the model that supports this equality.

As shown in Equation 1, the optimal Bayesian decision rule depends on the chosen loss function. In this example, we adopt the zero-one loss, which leads to predicting the label with the highest conditional probability. These probabilities are computed using the central distribution we previously derived.

To see how this approach works in practice, we carried out a synthetic experiment with the following setup. We considered binary attributes and generated each attribute independently with equal probability for 0 and 1. The class label was defined as a deterministic function of a fixed subset of 3 out of 12 total attributes, as shown in Table 1. The same subset of relevant attributes and labeling function were used across all trials.

Table 1: Mappings of 3-bit inputs under functions $f_1$ and $f_2$

| Input (bits) | $f_1$(Input) | $f_2$(Input) |
|---|---|---|
| 000 | a | A |
| 001 | b | B |
| 010 | c | A |
| 011 | d | C |
| 100 | e | D |
| 101 | f | D |
| 110 | g | D |
| 111 | h | E |

For each experiment, we generated a training set of size 20 and a test set of size 60. A larger test set was used to reduce variability in performance estimates, taking advantage of the synthetic nature of the data. To assess robustness to irrelevant features, we progressively removed one attribute at a time from the full set of 12, starting with irrelevant ones and stopping just before removing any of the three relevant attributes.

At each step, we trained and tested two versions of the central distribution: one assuming a deterministic label function (CD-Det) and one allowing for a conditional (non-deterministic) label distribution (CD-Prob). Both versions are constructed by aggregating over $2^{12}$ fused multinomial models, corresponding to all subsets of the 12 attributes. As a baseline, we also trained a decision tree classifier (DTree). All methods were trained on the same training data and evaluated on the same test data. Each configuration was repeated for 100 independent trials, and we reported the average accuracy (correct predictions divided by total test instances) over these trials.

The results, shown in Figures 1 and 2, reveal notable differences in performance depending on the underlying class assignment function. For function $f_1$, both deterministic and probabilistic methods achieve high accuracy, with the deterministic version of the central distribution slightly outperforming the other. In contrast, function $f_2$ is more difficult to predict, leading to a clearer separation in performance between the two methods. This difficulty primarily arises from strong non-deterministic dependencies between subsets of input bits and the assigned label. For instance, when the first input bit of

Figure 1: Accuracy vs. Number of Attributes (Function $f_1$, 20 train samples)

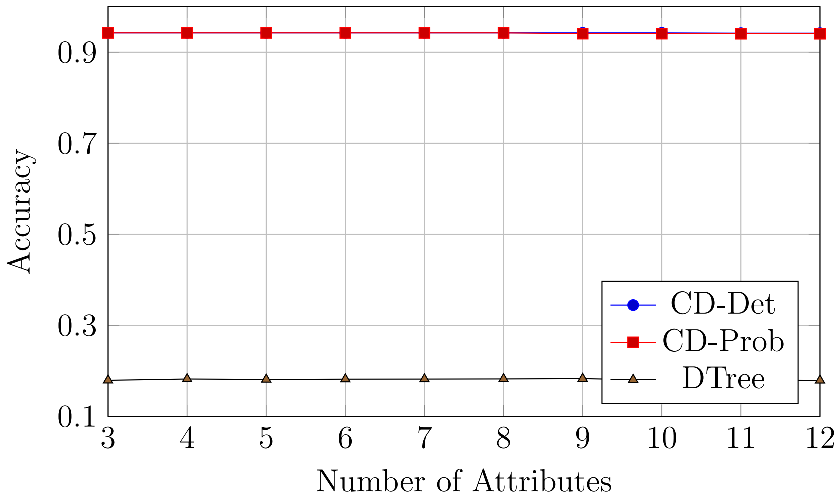
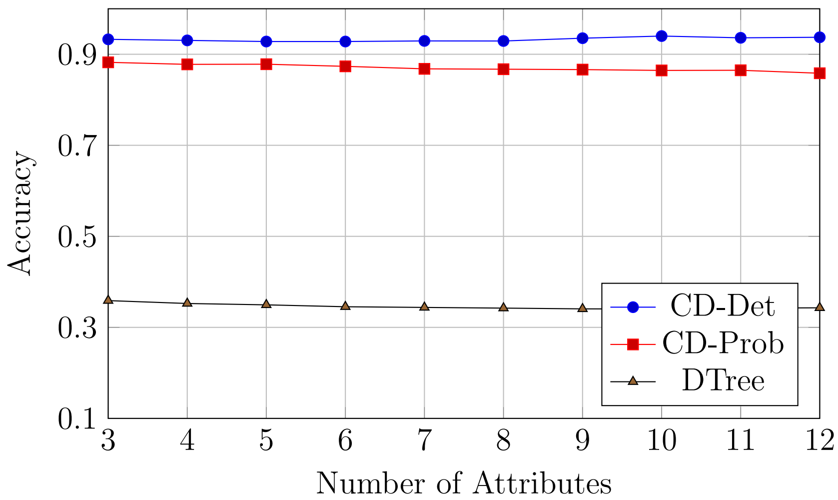
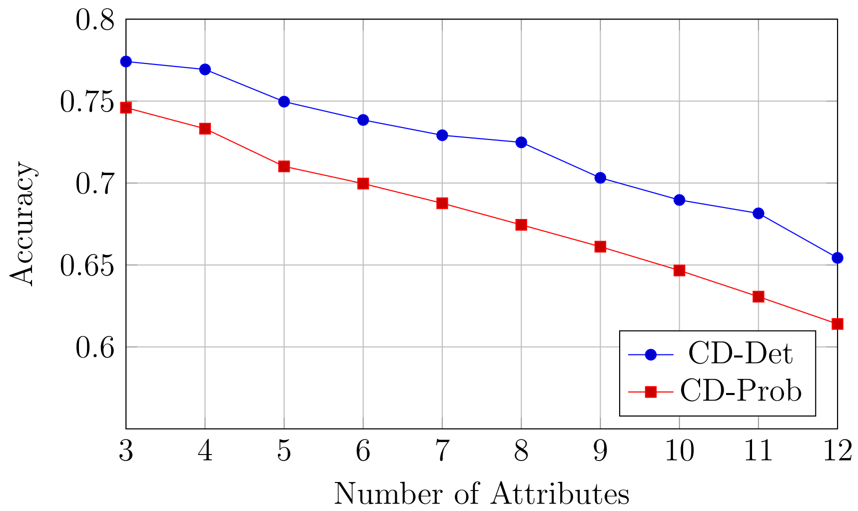



Figure 2: Accuracy vs. Number of Attributes (Function $f_2$, 20 train samples)



$f_2$ is 1, the label is often D, creating a misleading probabilistic association. In this case, the deterministic central distribution remains the most accurate overall, while the probabilistic version shows reduced performance due to misleading patterns.

Figure 3: Accuracy vs. Number of Attributes (Function $f_2$, 10 train samples)




While removing irrelevant attributes generally led to slight improvements in accuracy, the effect was not substantial in our primary experiments with 20 training samples. However, we observed that the benefit of attribute removal becomes more noticeable when the training set is smaller. In particular, additional experiments with 10 training samples showed a clearer performance gain as irrelevant features were removed, indicating that the impact of irrelevant attributes is more pronounced in low-data regimes (see Figure 3)

### 3.2 Latent Regression Model

We consider $C$ classes (or datasets), indexed by $c = 1, \ldots, C$. For each class $c$, we observe a data matrix $X^{(c)} \in \mathbb{R}^{N_c \times M}$, where each row $\boldsymbol{x}_i^{(c)} \in \mathbb{R}^M$ is associated with a $D$-dimensional latent variable $\boldsymbol{z}_i^{(c)} \in \mathbb{R}^D$.

Thus, each class is associated with a collection of latent variables

$$\boldsymbol{z}^{(c)} = \{\boldsymbol{z}_i^{(c)}\}_{i=1}^{N_c}, \qquad \boldsymbol{z}_i^{(c)} \in \mathbb{R}^D.$$

For each class $c$, the latent variables are assumed independent and identically distributed according to

$$\boldsymbol{z}_i^{(c)} \sim \mathcal{N}\left(\boldsymbol{\mu}^{(c)}, \tau_c^{-1} I_D\right), \qquad c = 2, \ldots, C,$$

where $\boldsymbol{\mu}^{(c)} \in \mathbb{R}^D$ is the class-specific latent mean and $\tau_c > 0$ is a precision parameter controlling the latent dispersion.

To remove redundancy, we fix the latent distribution of the first class as a reference coordinate system. Specifically, we impose

$$\boldsymbol{\mu}^{(1)} = \mathbf{0}, \qquad \tau_1 = 1,$$

so that

$$\boldsymbol{z}_i^{(1)} \sim \mathcal{N}(\mathbf{0}, I_D).$$

This anchoring removes global translation and scaling symmetries without restricting the expressive capacity of the model. The latent geometry of each class is therefore interpreted relative to the reference class $c = 1$.

Let $\phi : \mathbb{R}^D \to \mathbb{R}^K$ denote a fixed nonlinear basis mapping. The design matrix of class $c$ is defined as $Z^{(c)} \in \mathbb{R}^{N_c \times K}$ with entries

$$Z_{ik}^{(c)} = \phi_k(\boldsymbol{z}_i^{(c)}).$$

We collect the decoder weights in a shared matrix

$$W = [\boldsymbol{w}_1 \ \ldots \ \boldsymbol{w}_M] \in \mathbb{R}^{K \times M},$$

where each column $\boldsymbol{w}_j \in \mathbb{R}^K$ corresponds to the weights for the $j$-th observed dimension across all classes.

Conditioned on $\boldsymbol{z}^{(c)}$ and $W$, observations are generated as

$$p(X^{(c)} \mid \boldsymbol{z}^{(c)}, W) = \prod_{i=1}^{N_c} \prod_{j=1}^{M} \mathcal{N}\left( x_{ij}^{(c)} \,\middle|\, \phi(\boldsymbol{z}_i^{(c)})^\top \boldsymbol{w}_j, \beta_j^{-1} \right),$$

or equivalently,

$$\boldsymbol{x}_j^{(c)} \sim \mathcal{N}\left(Z^{(c)} \boldsymbol{w}_j, \beta_j^{-1} I_{N_c}\right),$$

where $\boldsymbol{x}_j^{(c)}$ denotes the $j$-th column of $X^{(c)}$ and $\beta_j > 0$ is the observation precision of that column.

To regularize the decoder, we assume noninformative priors for the latent parameters $\{\boldsymbol{\mu}^{(c)}, \tau_c\}_{c=2}^{C}$, and place independent Gaussian priors on the columns of $W$:

$$p(W) = \prod_{j=1}^{M} \mathcal{N}\left(\boldsymbol{w}_j \mid \mathbf{0}, \alpha_j^{-1} I_K\right).$$

The full joint distribution factorizes as

$$p(W) \prod_{c=1}^{C} p(X^{(c)} \mid \boldsymbol{z}^{(c)}, W) p(\boldsymbol{z}^{(c)} \mid \boldsymbol{\mu}^{(c)}, \tau_c) p(\boldsymbol{\mu}^{(c)}, \tau_c).$$

This defines a shared nonlinear generative model in which all classes are coupled through the common decoder $W$, while retaining class-specific latent geometry via $(\boldsymbol{\mu}^{(c)}, \tau_c)$.

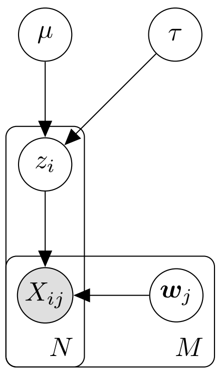



Figure 4: One class scalar latent regression model

### 3.2.1 Variational Inference for the Single-Class Case

To simplify the exposition, we now consider the single-class case. Since only one class is present, we remove the superscript $(c)$ from the notation for clarity.

Furthermore, we restrict the latent variable to be scalar,

$$\boldsymbol{z} = \{z_i\}_{i=1}^N \qquad z_i \in \mathbb{R},$$

so that

$$p(\boldsymbol{z} \mid \mu, \tau) = \prod_{i=1}^N \mathcal{N}(z_i \mid \mu, \tau^{-1}).$$

We treat $\mu$ and $\tau$ as unknown parameters to be inferred, using the standard joint Jeffreys prior for the mean and precision of a Gaussian model,

$$p(\mu, \tau) \propto \tau^{-1}.$$

The nonlinear basis mapping reduces to

$$\phi : \mathbb{R} \to \mathbb{R}^K,$$

and the design matrix $Z \in \mathbb{R}^{N \times K}$ is defined by

$$Z_{ik} = \phi_k(z_i).$$

We see the Bayesian network of this model in Figure 4. Exact posterior inference in this model is intractable due to the nonlinear dependence of the likelihood on $\boldsymbol{z}$. We therefore proceed by introducing a variational approximation.

In order to formulate a variational treatment of this model, we consider a variational distribution which factorizes between the latent variables and the parameters so that

$$q(\boldsymbol{z}, W, \mu, \tau) = q(\boldsymbol{z})q(W, \mu, \tau).$$

First, let us consider the derivation of the update equation for the factor $q(W, \mu, \tau)$. The log of the optimized factor is given by

$$\ln q^*(W, \mu, \tau) = \mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(W, \mu, \tau \mid \boldsymbol{z}, X)\right] + \text{const.}$$

As can be seen from Figure 4, the d-separation criterion implies that

$$\mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(W, \mu, \tau \mid \boldsymbol{z}, X)\right] = \mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(\mu, \tau \mid \boldsymbol{z})\right] + \sum_j \mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(\boldsymbol{w}_j \mid \boldsymbol{z}, X)\right].$$

Therefore, we have the following induced factorization for $q(W, \mu, \tau)$:

$$q(W, \mu, \tau) = q(\mu, \tau)\prod_{j=1}^{M} q(\boldsymbol{w}_j).$$

We obtain $q(\boldsymbol{w}_j)$ using the fact that each $\boldsymbol{w}_j$ is a weight vector of a Bayesian linear regression model:

$$\begin{aligned}
\ln p(\boldsymbol{w}_j \mid \boldsymbol{z}, X) &= -\frac{\beta_j}{2}\left\|Z\boldsymbol{w}_j - \boldsymbol{x}_j\right\|^2 - \frac{\alpha_j}{2}\boldsymbol{w}_j^\top \boldsymbol{w}_j + \text{const} \\
&= -\frac{1}{2}\boldsymbol{w}_j^\top(\beta_j Z^\top Z + \alpha_j I)\boldsymbol{w}_j + \beta_j \boldsymbol{x}_j^\top Z\boldsymbol{w}_j + \text{const.}
\end{aligned}$$

Taking the expectation with respect to $q(\boldsymbol{z})$, we obtain

$$\ln q^*(\boldsymbol{w}_j) = -\frac{1}{2}\boldsymbol{w}_j^\top(\beta_j \mathbb{E}\left[Z^\top Z\right] + \alpha_j I)\boldsymbol{w}_j + \beta_j \boldsymbol{w}_j^\top \mathbb{E}\left[Z\right]^\top \boldsymbol{x}_j + \text{const},$$

where expectations of matrices are taken element-wise. The resulting expression is quadratic in $\boldsymbol{w}_j$. Completing the square yields

$$q^*(\boldsymbol{w}_j) = \mathcal{N}(\boldsymbol{w}_j \mid \boldsymbol{\mu}_j, \Sigma_j).$$

The corresponding moments are

$$\begin{aligned}
\mathbb{E}\left[\boldsymbol{w}_j\right] &= \boldsymbol{\mu}_j, \\
\mathbb{E}\left[\boldsymbol{w}_j \boldsymbol{w}_j^\top\right] &= \boldsymbol{\mu}_j \boldsymbol{\mu}_j^\top + \Sigma_j,
\end{aligned}$$

with

$$\Sigma_j^{-1} = \beta_j \mathbb{E}\left[Z^\top Z\right] + \alpha_j I,$$
$$\boldsymbol{\mu}_j = \beta_j \Sigma_j \mathbb{E}\left[Z\right]^\top \boldsymbol{x}_j.$$

Since the design matrix satisfies $Z_{ik} = \phi_k(z_i)$, we have

$$\mathbb{E}\left[Z^\top Z\right] = \sum_{i=1}^{N} \mathbb{E}\left[\phi(z_i)\phi(z_i)^\top\right].$$

Collecting the posterior means into a matrix gives

$$\mathbb{E}\left[W\right] = [\boldsymbol{\mu}_1 \ \ldots \ \boldsymbol{\mu}_M].$$

Moreover, to simplify later expressions, define the diagonal precision matrix

$$B = \mathrm{diag}(\beta_1, \ldots, \beta_M).$$

Then, we have

$$\sum_{j=1}^{M} \beta_j \mathbb{E}\left[\boldsymbol{w}_j \boldsymbol{w}_j^\top\right] = \mathbb{E}\left[W\right] B \mathbb{E}\left[W\right]^\top + \sum_{j=1}^{M} \beta_j \Sigma_j.$$

This quantity will appear in the update for the latent variables. Note that if $\alpha_j = 0$ for all $j$, the mean simplifies to

$$\mathbb{E}\left[W\right] = \mathbb{E}\left[Z^\top Z\right]^{-1} \mathbb{E}\left[Z\right]^\top X.$$

We now consider the update equation for the factor $q(\mu, \tau)$:

$$\begin{aligned}
\ln q^*(\mu, \tau) &= \mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(\mu, \tau \mid \boldsymbol{z})\right] + \text{const} \\
&= -\frac{\tau}{2}\left(\sum_i \mathbb{E}\left[z_i^2\right] - 2\mu \sum_i \mathbb{E}\left[z_i\right] + N\mu^2\right) \\
&\quad + \left(\frac{N}{2} - 1\right) \ln \tau + \text{const}.
\end{aligned}$$

We define

$$\overline{z} = \frac{1}{N} \sum_i \mathbb{E}\left[z_i\right],$$
$$\overline{zz} = \frac{1}{N} \sum_i \mathbb{E}\left[z_i^2\right].$$

We observe that $q^*(\mu, \tau)$, for $N > 1$ follows a Normal–Inverse–Gamma distribution:

$$q^*(\mu, \tau) = \mathcal{N}\left(\mu \;\middle|\; \overline{z}, \frac{1}{N\tau}\right) \text{Gamma}\left(\tau \;\middle|\; \frac{N-1}{2}, \frac{N}{2}(\overline{zz} - \overline{z}^2)\right).$$

This implies

$$\begin{aligned} \mathbb{E}\left[\tau\right] &= \frac{(N-1)}{N(\overline{zz} - \overline{z}^2)}, \\ \mathbb{E}\left[\mu\tau\right] &= \overline{z}\mathbb{E}\left[\tau\right]. \end{aligned}$$

Finally, we derive the update equation for $q(\boldsymbol{z})$:

$$\begin{aligned} \ln q^*(\boldsymbol{z}) &= \mathbb{E}_{q(W,\mu,\tau)}\left[\ln p(X \mid \boldsymbol{z}, W) + \ln p(\boldsymbol{z} \mid \mu, \tau)\right] + \text{const} \\ &= \mathbb{E}_{q(W)}\left[\ln p(X \mid \boldsymbol{z}, W)\right] + \mathbb{E}_{q(\mu,\tau)}\left[\ln p(\boldsymbol{z} \mid \mu, \tau)\right] + \text{const}. \end{aligned}$$

We expand the terms $\ln p(X \mid \boldsymbol{z}, W)$ and $\ln p(\boldsymbol{z} \mid \mu, \tau)$. All expressions that are independent of $\boldsymbol{z}$ are treated as constants.

$$\begin{aligned} \ln p(X \mid \boldsymbol{z}, W) &= \sum_j \ln \mathcal{N}(\boldsymbol{x}_j \mid Z\boldsymbol{w}_j, \beta_j^{-1} I) \\ &= -\frac{1}{2}\sum_j \beta_j \left\|Z\boldsymbol{w}_j - \boldsymbol{x}_j\right\|^2 + \text{const} \\ &= \sum_i \left(-\frac{1}{2}\phi(z_i)^\top \sum_j \beta_j \boldsymbol{w}_j \boldsymbol{w}_j^\top \phi(z_i) + \phi(z_i)^\top \sum_j \beta_j X_{ij} \boldsymbol{w}_j\right) + \text{const}. \end{aligned}$$

$$\begin{aligned} \ln p(\boldsymbol{z} \mid \mu, \tau) &= -\frac{\tau}{2}\sum_i (z_i - \mu)^2 + \text{const} \\ &= \sum_i \left(-\frac{\tau}{2} z_i^2 + \tau\mu z_i\right) + \text{const}. \end{aligned}$$

Substituting these expansions into the variational update expression and taking the expectation yields:

$$\ln q^*(\boldsymbol{z}) = \sum_i \ln q^*(z_i),$$

where

$$\begin{aligned} \ln q^*(z_i) = &\left(-\frac{1}{2}\phi(z_i)^\top \sum_j \beta_j \mathbb{E}\left[\boldsymbol{w}_j \boldsymbol{w}_j^\top\right] \phi(z_i) + \phi(z_i)^\top \mathbb{E}\left[W\right] B X_{i,:}^\top\right) \\ &+ \left(-\frac{1}{2}\mathbb{E}\left[\tau\right] z_i^2 + \mathbb{E}\left[\tau\mu\right] z_i\right) + \text{const}. \end{aligned}$$

Here $X_{i,:}$ denotes the $i$-th row of the observation matrix $X$.

Because the update expression separates as a sum over $i$, the optimal factor $q(\boldsymbol{z})$ factorizes as $\prod_i q(z_i)$. Define

$$S^{-1} = \sum_j \beta_j \mathbb{E}\left[\boldsymbol{w}_j \boldsymbol{w}_j^\top\right],$$

$$\boldsymbol{m}_i = S\mathbb{E}\left[W\right] B X_{i,:}^\top .$$

Then, up to normalization,

$$q^*(z_i) \propto \mathcal{N}\left(\phi(z_i) \mid \boldsymbol{m}_i, S\right) \mathcal{N}\left(z_i \;\middle|\; \mathbb{E}\left[\mu\right], \mathbb{E}\left[\tau\right]^{-1}\right).$$

Note that since the basis mapping is nonlinear, this does not generally imply that $q^*(z_i)$ is Gaussian in $z_i$.

As seen previously, the update equations require expectations of the form

$$\mathbb{E}\left[\phi_i(z_k)\phi_j(z_k)\right], \quad \mathbb{E}\left[\phi_i(z_k)\right], \quad \mathbb{E}\left[z_k\right], \quad \mathbb{E}\left[z_k^2\right],$$

for all relevant indices $i, j, k$. These expectations typically do not admit closed forms and must be approximated numerically.

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